

Project 1 Plan / Computational Physics

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Overview

In this project I will write numerical integrators for two problems from classical mechanics. Both problems have an exact solution, so I can check my code against it. The first one is an ODE: two masses connected by springs. The second one is a definite integral: the distance travelled by an object that is slowing down because of drag.

(a) ODE: Two Mass Coupled Oscillator

I take two equal masses m that can slide on a track without friction. They are connected to each other and to two walls by three springs, all with the same stiffness k (wall- m_1 - m_2 -wall). I call x_1 and x_2 the displacements of the two masses from their equilibrium positions. Both masses only move along the same one spatial direction, but the system has two coordinates, so after I rewrite it as a first order system the state has four variables. The assignment lists “coupled oscillator” as an allowed example, I was thinking it as a great choice for part a which is an ODE.

To get the equations of motion I just look at which springs are pulling on each mass. The left spring pulls mass 1 back with a force $-kx_1$. The middle spring is stretched by an amount $x_2 - x_1$, so it pulls mass 1 with a force $+k(x_2 - x_1)$ and mass 2 with the opposite force $-k(x_2 - x_1)$. The right spring pulls mass 2 back with $-kx_2$. Putting these into Newton’s second law I get

$$m\ddot{x}_1 = -kx_1 + k(x_2 - x_1) = -2kx_1 + kx_2, \quad (1)$$

$$m\ddot{x}_2 = -k(x_2 - x_1) - kx_2 = kx_1 - 2kx_2. \quad (2)$$

These are two second order equations, but the integrators I will write need a first order system. So I define the state $\mathbf{y} = (x_1, x_2, v_1, v_2)$ and use $\dot{x}_i = v_i$ together with the two equations above.

To find the normal modes I try solutions of the form $x_1, x_2 \propto e^{i\omega t}$, which turns the system into a small eigenvalue problem. Solving it gives two frequencies,

$$\omega_+ = \sqrt{k/m} \quad (\text{in-phase}), \quad \omega_- = \sqrt{3k/m} \quad (\text{out-of-phase}). \quad (3)$$

The in-phase mode is the one where the two masses move together ($x_1 = x_2$), so the middle spring is never stretched and only the two outer springs act that is why the frequency is just $\sqrt{k/m}$. The out-of-phase mode is when they move against each other ($x_1 = -x_2$); now the middle spring is stretched twice as much, so the effective spring constant is bigger and the frequency is higher. Any initial condition can be written as a combination of these two modes, so I have a closed form solution for $x_1(t)$ and $x_2(t)$ that I will use to check the integrators.

Physical behaviors to verify:

1. **Energy conservation.** The total energy is the sum of the kinetic energy of the two masses and the potential energy stored in the three springs (each spring contributes $\frac{1}{2}k(\Delta\ell)^2$, where $\Delta\ell$ is how much it is stretched):

$$E = \frac{1}{2}m(v_1^2 + v_2^2) + \frac{1}{2}k[x_1^2 + (x_2 - x_1)^2 + x_2^2].$$

Since there is no friction, E should stay constant in time. For explicit Euler applied to an oscillator the numerical energy does not just drift randomly; it typically grows with time, so I expect to see the amplitude of the motion getting bigger and bigger as the simulation runs. RK4 does much better in practice, but it also does not conserve energy exactly, so I still expect a small slow drift there. This is similar to what we saw in the in-class spring oscillator exercise, where even RK2 was already much closer to the exact solution than Euler.

2. **Normal-mode purity and beating.** If I start the system in a pure mode ($x_1 = x_2$ or $x_1 = -x_2$, both masses at rest), it should stay in that mode and oscillate at the corresponding $\omega_{\pm}$. If instead I only move one mass at the beginning, I excite both modes equally, so the solution is a sum $\cos(\omega_+t) + \cos(\omega_-t)$. Using the identity

$$\cos(\omega_+t) + \cos(\omega_-t) = 2 \cos\left(\frac{\omega_- - \omega_+}{2}t\right) \cos\left(\frac{\omega_- + \omega_+}{2}t\right),$$

this looks like a fast oscillation at the average frequency $(\omega_- + \omega_+)/2$ inside a slow envelope at $(\omega_- - \omega_+)/2$. So the beating is governed by the difference between the two normal-mode frequencies, and I should see energy going back and forth between the two masses on that slower timescale.

(b) Definite Integral: Stopping Distance Under Linear Drag

Now I take an object of mass m moving in one dimension with initial speed v_0 , with only a linear drag force $F = -bv$ acting on it (the minus sign is because drag always pushes against the motion).

From Newton's second law, $m\dot{v} = -bv$, which I can rewrite as

$$\frac{dv}{v} = -\frac{b}{m} dt.$$

Integrating both sides gives $\ln v = -bt/m + \text{const}$, and using $v(0) = v_0$ I get

$$v(t) = v_0 e^{-t/\tau}, \quad \tau = m/b, \tag{4}$$

where τ is the characteristic time it takes for the speed to drop by a factor of e . Since velocity is dx/dt , the distance travelled between $t = 0$ and $t = T$ is

$$x(T) = \int_0^T v_0 e^{-t/\tau} dt, \tag{5}$$

and doing this integral by hand gives

$$x(T) = v_0 \tau (1 - e^{-T/\tau}). \tag{6}$$

Physical behaviors to verify:

1. **Short-time limit** ($T \ll \tau$). For small T the drag has not done much yet, so the object should move almost like a free object, and $x(T) \rightarrow v_0 T$. I can check this directly from the exact solution by expanding $e^{-T/\tau} \approx 1 - T/\tau$, which gives $x(T) \approx v_0 T$.
2. **Total stopping distance** ($T \rightarrow \infty$). The object never really stops in finite time, but the total distance it can travel is finite. Taking $T \rightarrow \infty$ in the exact solution kills the exponential and I am left with $x_\infty = v_0 \tau$.